

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Electrical System Sizing Calculation
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Electrical System Sizing Calculation

R01	08.08.22	Issued for Review	N. Kari	V. Ani	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	08.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001	Electrical Load List
EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-015	Overall Key Single Line Diagram Power Supply
EA-BFPCL-GPMC-GGPL-FEED-ELE-SPC-002	Specifications for Electrical Equipment
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AC	Alternating Current
DB	Distribution Board
DC	Direct Current
EPC	Engineering, Procurement and Construction
EMC	Electromagnetic Compatibility
IEC	International Electro Technical Commission
IEEE	Institute of Electrical and Electronics Engineers
IFC	Issued for Construction
IFD	Issued for Design
IP	Ingress Protection
ISO	International Organisation for Standardization
LV	Low Voltage
MTO	Materials Take Off
PCS	Process Control System
PFD	Process Flow Diagram
P&ID	Piping and Instrumentation Diagram
UPS	Uninterruptible Power Supply
XPLE	Cross-Linked Polyethylene

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

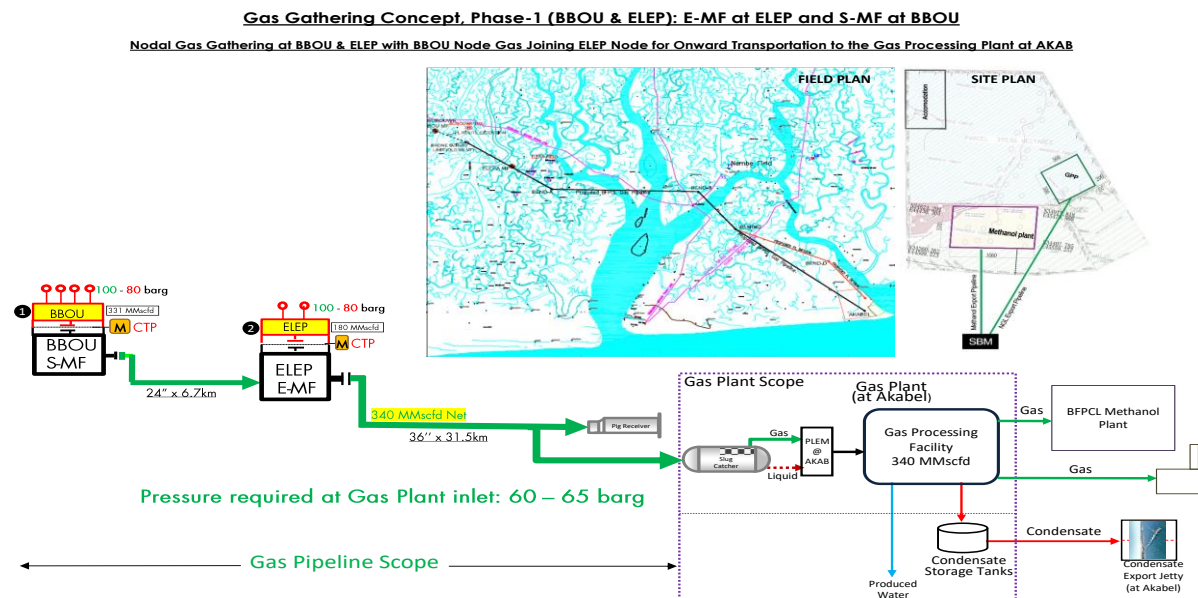


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The scope of this report is to perform calculations for the sizing and selection of Low Voltage Switchgear specified on the Overall Single Line Diagrams and Technical Data Sheet (TDS), obtainable as part of this project's scope and step-down Transformer.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
BS 5486	Specification of low voltage switchgear and control gear assemblies
BS 4752	Specification for switchgear and control gear for voltages up to and including 1000Vac and 1200Vdc circuit breaker
IEC 60947	Low-voltage switchgear and control gear
IEC 60269	Low-voltage fuses
IEC 60947-2	Low-voltage switchgear and Control gear. Part 2: Circuit breakers
IEC 60947-3	Low voltage switchgear and control gear - Part 3: Switches, disconnectors, switch-disconnectors and fuse-combination units.
IEC 60947-4-1	Low voltage switchgear and control gear - Part 4: Contactors and motor starters - Section one - Electromechanical contactors and motor starters.
IEC 60364-1	Low voltage electrical installations; part 1; gen characteristics
IEC 60934	Circuit-breakers for equipment (CBE)
IEC 60529	Classification of Degrees of Protection Provided by Enclosures
IEC 60044-1	Instrument Transformers - Current transformers
IEC 61641	Enclosed low voltage switchgear & control gear Guide for testing under conditions of arcing due to internal fault
IEC 62053-11	Electrical metering Equipment; Class 0.5, 1 and 2 alternating-current watt-hour meters
IEC 61810-2	Electrical metering Equipment; Class 0.5, 1 and 2 alternating-current watt-hour meters

IEC 61811-10	Electromechanical elementary relays; part 10 specification
IEC 61811-11	Electromechanical elementary relays; part 11 specification
IEC 60304	Standard Colours for Insulation for Low-Frequency Cables and Wire
IEC 60721-2-1	Classification of Environmental Conditions: Part 2-1:
IEC 61554	Dimensions for panel-mounted indicating and recording electrical measuring instruments
IEC 61508	Safe Capability: system related with safety
IEC 60079	Electrical apparatus for explosive gas atmospheres (part 10 & 14)
IEC 61779	Electrical apparatus for the detection and measurement of flammable gases
IEC 60099	Surge arresters and control gear standards
IEC 61439-1	Low-voltage switchgear and control gear assemblies
IEC 60255	Electrical relays
IEC 62271	Standard for high-voltage switchgear and Control gear
IEC 60073	Basic and safety principles for man-machine interface, marking and identification - Coding principles for indicators and actuator
IEC 62353	Electrical safety testing
IEC 60185	Current Transformers
ISO 9001:2000	Quality Management Systems-Requirements
IEEE C37.2	ANSI/IEEE Standard Device Numbers
UI 508A	Standard for Industrial Control Equipment Health and Safety at Work Act, 1974 Section 6.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Transformer Sizing Methodology

3.1. Low Voltage 11/0.4kV Step-down Transformer

From load list document number EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001

SUMMARY - OVERALL LOAD			
	kW	kVA _r	kVA
Normal Running Load (X%TC + Y%TI)	17.44	13.08	21.80
Peak Load (X%TC + Y%TI + Z%TS)	17.55	13.17	21.94
Where: C = Continuous Load I = Intermittent Load S = Standby Load TC = Total Continuous Load TI = Total Intermittent Load TS = Total Standby Load X% = 100% Y% = 30% Z% = 10% V = Vital Load E = Emergency Load N = Normal Load			

SUMMARY

Power Requirement: -	18	kW
	22	kVA
Power Requirement including 25% margin	23	kW
	27	kVA
Recommended Gas Engine Gen. Size	50	kVA

Therefore, recommended standardized Transformer size is 50kVA, 400V, 50Hz

One 50KVA KNAN DYn11 11/0.4kV step-down is provided, a second unit of similar size can be provided as spare to satisfy n+1 sparing philosophy.

4. SwitchGear Ampacity Calculation Methodology

4.1. LV Switchgear (SB-01)

Switchgear ampacity bus rating shall be based on 1 no. 50kVA connected Transformer (see single line diagram drawing number EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001). Meanwhile, the switchgear short circuit current rating shall also be based on one connected transformer in operation as well as connected motor fault level contributions.

Per section 4.1 above, Transformer power rating = 50kVA

Power equation,

$$S = \sqrt{3} V_L I \text{ kVA}$$

Where,

I = ampacity (A)

S = Apparent power (kVA)

V_L = Generator terminal line Voltage

From the Power equation above

$$I = 1000 * S / (\sqrt{3} * V_L)$$

By substitution for S and V_L in power equation above;

$$I = 1000 * 50 /$$

$$I = 72.17A$$

Hence, recommended switchgear breaker & bus rating sizing is 120A.

5. UPS Sizing Methodology

5.1. Design Note and Assumption

- i. Manifold Station UPS system 230Vac system autonomy time is 4 hours' back-up each.
- ii. Charger shall be fully rated for battery charging and supply power to the connected load through the static inverter simultaneously.
- iii. Power factor for UPS inverter shall be used as 0.92 lag.
- iv. Battery aging factor is 20% as per IEC 60623, DC supply.,
- v. Battery design margin is 10% as per IEC 60623.
- vi. Battery sizing calculation is as per IEEE STD 1115.
- vii. Batteries shall be nickel-cadmium or lithium ion phosphate type.
- viii. UPS load is contained in the overall load balance list.

Cable sizing for all cable related items is per OHF-NET-OHSF1-L44-00003

5.2. UPS Sizing Calculation

The total load fed from the UPS (UPS-01), referring to the load list,

UPS efficiency at 92%

Maximum load with 25% spare capacity = 4.0KW

Maximum load without 25% spare capacity = 3.0KW

Total load with 25% spare capacity= 4.0KW

$$KVA = 4/0.92 = 4.35KVA$$

The selected UPS shall be standard rating 5kVA.

Switchboard voltage rating at 400V 0.8 p.f;

Maximum Bus current rating for this UPS distribution board output is;

$$A = 5 \times 1000$$

$$400 \times 0.8$$

$$A = 15.62A$$

The standard current rating is selected to be 32A, Therefore, UPS System is rated to be 5KVA, 400/230VAC 3phase incomer, single phase outgoer UPS.

